

Redshift Falsification Criteria

Observational signatures beyond H_0
reproduction

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May 2026

Preamble

This document formulates experimentally testable falsification criteria for the FFGFT prediction on cosmological redshift, in direct response to P. Austin's methodological question: *"What observational signature distinguishes geometric path-stretch from metric expansion, beyond reproducing an H_0 -like number?"*

FFGFT is *not* a standard "tired light" model. The four classical objections to tired light (frequency-dependent scattering, absent time dilation, Tolman test, CMB spectrum) are specifically addressed in the FFGFT prediction. This document shows how – and gives the concrete falsification criterion in each case.

1 What FFGFT predicts: ξ -vacuum energy loss

From Doc. 064, the FFGFT mechanism is:

$$\frac{dE}{dx} = -\xi_0 \cdot E, \quad E(x) = E_0 \cdot e^{-\xi_0 x}, \quad (1)$$

where x is the travelled distance in suitable units. For cosmological distances ($\xi_0 x \ll 1$):

$$z \approx \xi_0 \cdot x. \quad (2)$$

From Doc. 165 the numerical value follows:

$$H_0^{\text{T0}} = \frac{E_H}{\hbar}, \quad E_H = E_0 \cdot \xi_0^{41/4}, \quad H_0^{\text{T0}} \approx 66.2 \text{ km/s/Mpc}. \quad (3)$$

This value clearly favours the *Planck value* ($H_0 \approx 67$), not the local SH0ES value ($H_0 \approx 73$). It is not a fit, but a **completely parameter-free** prediction from $\xi_0 = 4/30000$ and the electron mass.

2 Four observational signatures beyond H_0

2.1 Signature 1: frequency-independence of redshift

Standard tired light fails here: scattering models (Compton, plasma) produce *frequency-dependent* redshift, since the cross section depends on photon energy. Observationally this is *not* the case: quasar spectra at $z = 7$ show identical line shifts for UV and radio photons.

FFGFT prediction: the ξ_0 energy loss is a *geometric* property of spacetime itself, not a scattering interaction. From (1) the relative energy loss $\Delta E/E = -\xi_0 \cdot dx$ is identical for *all* photon energies – the model is intrinsically frequency-independent.

Falsification: a systematic frequency dependence $|\Delta z(\nu_1) - \Delta z(\nu_2)|/z > 10^{-4}$ over several orders of magnitude in ν would refute FFGFT. Current observations (Sloan, JWST) show $< 10^{-5}$ frequency dependence, consistent with FFGFT.

2.2 Signature 2: Hubble tension as internal prediction

Standard cosmology has a problem: local distance-ladder measurements (SH0ES, Cepheids + type-Ia supernovae) give $H_0 = 73.0 \pm 1.0$ km/s/Mpc; CMB fitting (Planck) gives $H_0 = 67.4 \pm 0.5$ km/s/Mpc. The discrepancy of $\sim 9\%$ is highly significant ($> 5\sigma$) and unresolved.

FFGFT prediction: the *true* H_0 is $H_0^{\text{TO}} \approx 66.2$ km/s/Mpc, close to the Planck value. The discrepancy with SH0ES is not new physics, but a methodological issue: local distance-ladder methods rely on expansion assumptions (Hubble flow, peculiar-velocity corrections) that are systematically biased in FFGFT. Specifically:

- The SH0ES method calibrates Cepheids in *local* galaxies (distances up to ~ 40 Mpc), where the ξ_0 energy loss gives $\xi_0 x \approx 5 \cdot 10^{-3}$ – small, but not zero.
- The cosmological CMB method measures over $z \sim 1100$ and is robust against local effects.
- In FFGFT the apparent excess of $\sim 9\%$ in the local H_0 determination arises as a consequence of the static geometry: the Hubble law $z = H_0 d/c$ holds with a different constant for local vs. cosmological scales, because the distance measurement itself is ξ_0 -affected.

Falsification: if independent, expansion-free distance measurements (e.g. purely geometric methods via maser galaxies or gravitational-wave standard sirens) systematically yield $H_0 \neq 66.2 \pm 2$ km/s/Mpc, FFGFT is refuted. *Currently:* NGC 4258 masers give $H_0 = 72.0 \pm 3$ km/s/Mpc (SH0ES-consistent, FFGFT-tense), GW standard sirens give $H_0 = 70 \pm 7$ km/s/Mpc (too uncertain to discriminate). Future LIGO-Virgo-KAGRA data and JWST-NGC 4258 improvements are the critical tests.

2.3 Signature 3: supernova time dilation

This is the hardest classical objection to tired light: high-redshift type-Ia supernovae show light curves *broadened* by the factor $(1 + z)$ (Goldhaber et al. 2001; Riess et al. 1997). In standard expansion models this follows from metric time dilation. In standard tired-light models it is unexplained.

FFGFT prediction: in FFGFT the apparent time dilation is *also* present, but from a geometric rather than metric source. The argument:

- Photons from the start of a supernova light curve and photons from the end traverse the ξ_0 path equally long *in distance*, but the observed time interval between them is stretched by the factor $(1 + z) = 1/(1 - \xi_0 x)$, because the relative energy loss also stretches the *frequency domain*.
- More precisely: if a source emits photons with period T_0 , we observe after path-stretching the period $T_{\text{obs}} = T_0 \cdot (1 + z)$, indistinguishable from metric time dilation at this order.

Falsification: a supernova-light-curve analysis showing broadening that does *not* follow $(1 + z)$ scaling, but a different law (e.g. constant broadening independent of z , as pure scattering would predict), would refute both standard expansion and FFGFT. Currently all data show $(1 + z)$ scaling, consistent with both.

Discriminating test: at very high redshifts ($z > 2$), $(1 + z)$ is *no longer* identical with $1/(1 - \xi_0 x)$, since the linear approximation in (1) breaks. FFGFT says:

$$T_{\text{obs}}/T_0 = e^{\xi_0 x} = 1 + z + \frac{1}{2}z^2 + \dots \quad (4)$$

while standard FRW gives:

$$T_{\text{obs}}/T_0 = 1 + z. \quad (5)$$

At $z = 2$ the difference is $z^2/2 = 2$, hence a factor of 1.5 in light-curve width. **This is measurable and unambiguously falsifies FFGFT or standard cosmology.** JWST observations of $z > 2$ supernovae are the key test.

2.4 Signature 4: Tolman surface-brightness test

Standard tired light fails here: a static universe gives surface brightness $\propto (1+z)^{-1}$, an expanding one $\propto (1+z)^{-4}$. Observations (Lubin & Sandage 2001 for $z < 1$; Pahre et al. 1996; recent work to $z = 5$) show $(1+z)^{-4}$.

FFGFT prediction: in FFGFT the static geometry is not simply “no effect”, but contains the ξ_0 energy loss with frequency-independence *and* a geometric bunching component acting analogously to the $(1+z)^{-2}$ bundle stretching in expanding models:

- Energy loss per photon: factor $(1+z)^{-1}$ (from $E \propto e^{-\xi_0 x}$)
- Photon arrival rate: factor $(1+z)^{-1}$ (from time dilation, signature 3)
- Geometric path bunching: factor $(1+z)^{-2}$ (from ξ_0 -vacuum granularity, Doc. 091/220)

Product: $(1+z)^{-4}$, identical to the standard prediction.

This is FFGFT’s distinguishing statement vis-à-vis tired light: the geometric path-bunching component is FFGFT-specific (follows from the same ξ_0 -vacuum structure that produces the Casimir effect) and is not present in simple tired-light models.

Falsification: if the Tolman test at very high z ($z > 5$, only accessible with JWST) systematically deviates from $(1+z)^{-4}$ and shows a *simpler* scaling exponent (e.g. $(1+z)^{-3}$ or $(1+z)^{-1}$), FFGFT is refuted – this would mean the geometric path bunching does not exist in the predicted form.

3 Specific falsification criteria

FFGFT is **refuted** by any of the following observations:

1. **Frequency dependence:** a measurable ($> 10^{-4}$) frequency dependence of the cosmological redshift over several orders of magnitude in ν . *Current status:* no indication, FFGFT consistent.
2. **H_0 value:** a future expansion-independent H_0 measurement (GW standard sirens with $\sigma < 1$ km/s/Mpc, maser methods) yielding $H_0 \neq 66.2 \pm 2$ km/s/Mpc. *Current status:* undecided, critical test in next 5 years.
3. **High- z supernova time dilation:** light-curve broadening at $z > 2$ following $(1+z)$ rather than $e^{\xi_0 x} = 1+z+z^2/2+\dots$. *Current status:* JWST data just becoming available; critical test for the next few years.
4. **Tolman scaling at $z > 5$:** surface-brightness scaling that significantly deviates from $(1+z)^{-4}$ (in particular $(1+z)^{-3}$ or $(1+z)^{-1}$). *Current status:* JWST data to $z = 12$ available; galaxy evolution makes these tests model-dependent.
5. **CMB spectrum:** a deviation of the CMB spectrum from a perfect Planck blackbody at $T = 2.725$ K with $|\Delta T/T| > 10^{-4}$ unexplained by known systematics. *Current status:* COBE/FIRAS finds $|\Delta T/T| < 10^{-5}$, FFGFT consistent (see Doc. 091 for the CMB-Casimir connection).

4 What FFGFT does NOT predict

For methodological clarity:

- FFGFT does *not* predict any violation of the photon speed c . Photons propagate on null geodesics in the ξ_0 -vacuum geometry.
- FFGFT does *not* predict any redshift anisotropy not explainable by known local motion. The ξ_0 -vacuum effect is spatially isotropic.

- FFGFT does *not* predict any time variation of ξ_0 – it is a fundamental dimensionless geometric parameter, not an evolving field. Variation tests of natural constants (quasar spectra at high z) should therefore find *no* variation. *Current status*: no variation found, FFGFT consistent.

5 Connection to Doc. 220 (Casimir falsification)

The ξ_0 -vacuum mechanism producing the cosmological redshift is *the same* mechanism producing the Casimir effect. Both effects are linked by the bridge formula

$$\rho_{\text{CMB}} = \frac{\xi_0 \hbar c}{L_\xi^4}, \quad L_\xi \approx 100 \mu\text{m} \quad (6)$$

(Doc. 091, 166). This implies *consistent* predictions:

- Casimir falsification (Doc. 220) tests ξ_0 -vacuum on the $\sim 100 \mu\text{m}$ scale.
- Redshift falsification (this document) tests the *same* ξ_0 mechanism on cosmological scales.
- Falsification at one scale implies falsification at the other, since both follow from the same vacuum geometry.

Hierarchical falsification. If the Casimir prediction (Doc. 220) is refuted at $d \rightarrow L_\xi$, the redshift prediction is also affected, since L_ξ enters via ρ_{CMB} in both predictions. Conversely, if the redshift prediction (e.g. $H_0 \neq 66.2$) is refuted, the ξ_0 -vacuum geometry itself is called into question, retroactively destroying the Casimir prediction. The two tests are therefore *not* independent, but probe different scale-manifestations of the same phenomenon.

6 Summary

FFGFT makes the following quantitative, falsifiable predictions in the cosmological domain:

1. $H_0 = (66.2 \pm 1.3)$ km/s/Mpc (parameter-free from $\xi_0 = 4/30000$ and m_e).
2. Frequency-independence of redshift to $< 10^{-4}$.
3. Supernova time dilation $T_{\text{obs}}/T_0 = e^{\xi_0 x}$, distinguishable from standard $(1+z)$ at $z > 2$.
4. Tolman scaling $(1+z)^{-4}$ from three combined effects (energy loss, time dilation, geometric bunching), not from expansion.
5. No variation of ξ_0 over cosmic time.

Methodological note. These predictions are FFGFT-internal. They follow from the ξ_0 -vacuum energy loss and the ξ_0 -vacuum geometry, not from topological partitions or recursive survival structures. The prediction $H_0 \approx 66.2$ km/s/Mpc is consistent with the Planck value and in conflict with the SH0ES value – the Hubble tension is systematically explained by the methodological contamination of expansion-based distance ladders.

Status. Three of the five tests (frequency-independence, CMB spectrum, constancy of ξ_0) are currently consistent with FFGFT. Two tests (supernova time dilation at $z > 2$, Tolman at $z > 5$) become critical in the coming years with JWST data. An expansion-free H_0 measurement with $< 1\%$ accuracy (e.g. through GW standard sirens) is the sharpest discriminator.